For frist week , just understand the code base and run the kinematic gait in sim

Was able to produce static trajectory but lot of problem in doctor and build

Move onto getting Would also need to know deflection range and stiffness.Figure out if it is possible to get joint torque profiles from simulation for kinematic gaits

Got the deflection range from the git hex file

In week 3 Reviewed preliminary results for the robot. The produced graphs of torque appear random, possibly something related to the kinematic controllers. And the actuator isnt reaching the commanded pos

Depper analysis finds:
The surface coefficient has been set to 0.7, which is correct, but the contact stiffness has not been set. Gazebo has a high default value for this.More torque is being taken by hips, which is to be expected, and discussed the fact that the PID is the same for every PWM.Possible reasons for the issues discussed include the fact that the legs are dragging while moving and/or that the center of mass changes when a leg lifts up which has not been taken into account.May have to design individual gaits to fix the leg dragging problem.

Decrease contact stiffness and run the simulation again and verify results.

what force torque is sensing (contorl , effort or troque

another way to put forrce tsnesor foot and usse jacobian for linking to joint

pos to torque paper (mentioned in thesis)

filter out the signal

add the block and put robot on top and run gati(starting point)

early works on dynamical model of quadruped

underactuated

The above things are mostly minutes of meeting with prof
Basic need I set up this which is for basically Hyderabad simulation in rows p
Initially setu the codebase understand the scripts what it how its running gaits cycle everything
But the result were very erratic random , I analyzed and figure these coud be potential problem
No joint damping/friction,No foot contact model/friction
Untuned, uniform PID gains

But even after tuning this toque didnt change much ,but it was improvement on states , the actuator were closely following the commandsbut then professor notice that we set the forque torqu at 50hz but sampling at 10hz we correct them initially by correcting the logging system but then there was problem its till only log at 5hz but copy over as sim was 11 time slower even though we set 1 factor we were logging inreal time so cache update aevery 5 sample so we then switch to sim clock now torque result are good as actuator have time inbetween to move between command
Now we have move to dynamic model, reading book , foundation of model and underactuated robots for dynamics model so next sstep is work on dynamic model of robot